

Time Series Project:

ARIMA Modelling of a French Industrial Production Index

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Part I – The Data

Description of the Series

The series under analysis is the seasonally and working-day adjusted Industrial Production Index (IPI) for the pharmaceutical industry in metropolitan France, published by the French national statistics office (INSEE). This monthly series contains over 400 observations, covering the period from January 1990 to February 2025, and is expressed in base 100 as of January 2021.

The series displays a highly non-linear trend. From the early 1990s to the early 2010s, it shows moderate growth with notable fluctuations, followed by a strong upward trend from around 2015 onward. A particularly sharp increase is observed after 2020, possibly reflecting the pandemic-driven surge in pharmaceutical activity, notably related to COVID-19 vaccine production and increased demand for health-related products.

As shown in Figure 1, the series exhibits non-constant variance and a visibly increasing mean. These characteristics suggest non-stationarity, which will be confirmed and addressed in the following section. A number of outliers can also be observed, particularly around recent peaks, potentially corresponding to specific economic shocks or industry-specific events.

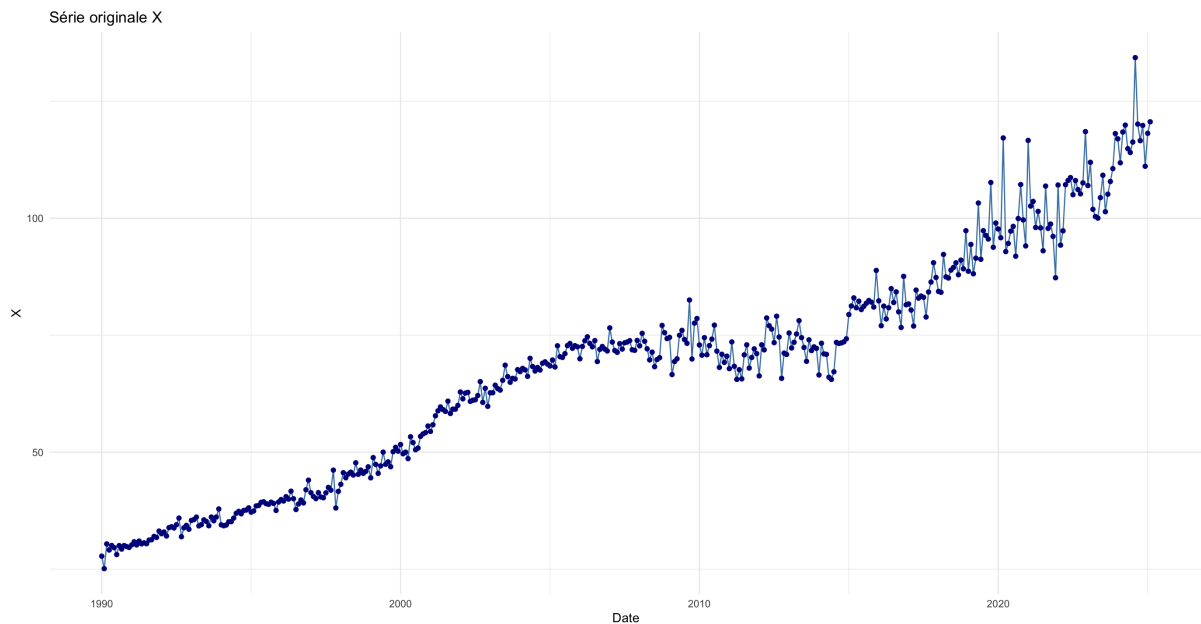


Figure 1: Original IPI series for the pharmaceutical sector (CVS-CJO, Jan. 1990–Dec. 2024).

Due to the presence of an upward trend and heteroskedasticity, a transformation is required in order to model the series under the assumption of stationarity. In the following sections, we will explore logarithmic and differencing techniques to achieve this objective.

Analysis of Series Stationarity

To formally assess the stationarity of the original time series, we conducted two widely-used statistical tests: the Augmented Dickey-Fuller (ADF) test and the Phillips-Perron (PP) test.

For the ADF test, we determined the appropriate number of lags by performing Ljung-Box Q-tests on the residuals. This approach ensured that the residuals exhibited no significant autocorrelation, leading us to select 21 lags. The resulting test statistic was -0.7832, accompanied by a p-value of 0.9628, indicating a failure to reject the null hypothesis of a unit root. In contrast, the Phillips-Perron test yielded a statistic of -95.411, with a p-value less than 0.01. This might suggest stationarity at first glance, but given the inconsistency with visual inspection, the result appeared potentially misleading.

To resolve this discrepancy, we applied a first difference transformation defined as $X_t - X_{t-1}$. The results following this transformation were more consistent across both tests. The ADF test statistic improved significantly to **-4.9194**, with a corresponding **p-value of 0.01**. The PP test produced a much lower statistic of **-524.68**, again with a **p-value below 0.01**, supporting the presence of stationarity.

Despite these encouraging results, a visual inspection of the differenced series (see Figure 2) revealed signs of variance instability. To quantify this, we divided the series into three equal segments and computed the variance within each. The first third exhibited a variance of 2.87, the second third increased to 9.85, and the final third reached 45.34. This clear upward trend in variance suggested the need for a more robust transformation.

Consequently, we applied a logarithmic transformation to stabilize the variance, followed by first differencing to address potential non-stationarity. This transformation, expressed as $\log(X_t) - \log(X_{t-1})$, provided strong statistical evidence of stationarity. The ADF test produced a statistic of **-6.7006** with a **p-value of 0.01**, while the PP test returned a statistic of **-525.85** with a **p-value below 0.01**.

In addition to statistical confirmation, we also observed that the variance in the log-differenced series remained relatively stable across time. The first third showed a variance of 0.0018, the second third measured at 0.0019, and the final third rose only slightly to 0.0046. These findings justified the use of the log-differenced series in subsequent modeling stages.

To streamline and formalize our testing process, we developed a set of functions in R. These included an ADF testing procedure with automatic lag selection based on Ljung-Box diagnostics to ensure well-specified residuals. Additionally, we implemented the Phillips-Perron test to corroborate findings from the ADF approach. These automated procedures provided a robust and replicable methodology for assessing the stationarity of various transformed time series.

Graphical Representations

We provided visualizations of:

- The original time series
- The differenced series
- The log-differenced series

These representations allow visual inspection of trends and variance stability.

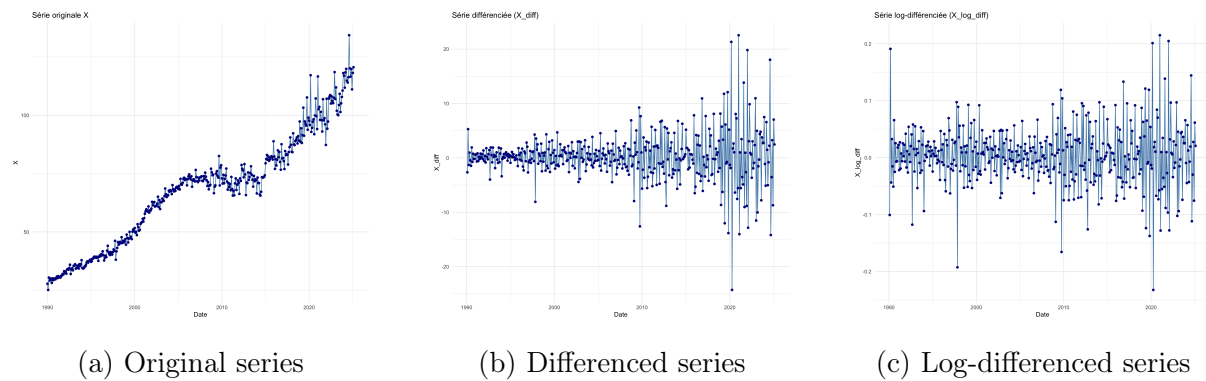


Figure 2: Visual comparison of the original, differenced, and log-differenced versions of the IPI series.

Part II – ARMA Modelling

ARMA Model Selection and Diagnostic

After applying a logarithmic transformation followed by first-order differencing, the original time series became stationary. To guide $\text{ARMA}(p, q)$ model identification, we examined the autocorrelation (ACF) and partial autocorrelation (PACF) plots in Figure 3.

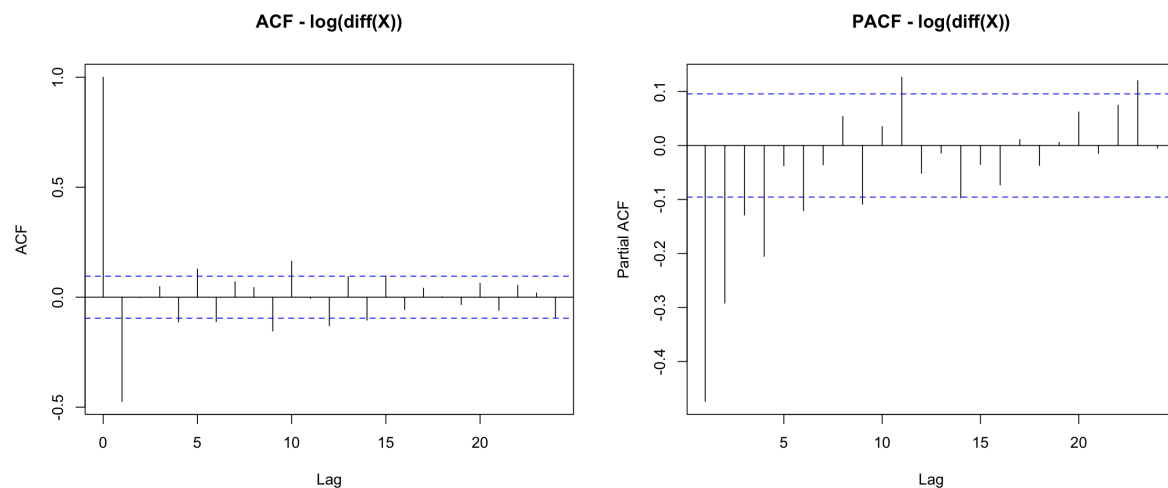


Figure 3: ACF and PACF of the log-differenced series

The ACF drops sharply after lag 1 and exhibits smaller oscillations up to lag 10, while the PACF shows a gradual decay with notable spikes beyond lag 1. This pattern suggests both autoregressive and moving average components, supporting the use of an ARMA model. The ACF cutoff implies a low-order MA (likely order 1 or 2) and the trailing PACF implies an AR structure.

To select an appropriate model, we estimated all $\text{ARMA}(p, q)$ combinations with $p, q \in \{0, 1, 2, 3\}$, and evaluated them using the Akaike Information Criterion (AIC) and the Bayesian

Information Criterion (BIC). AIC favored a more complex ARMA(3,3) model ($AIC = -1452.81$), while BIC, which penalizes complexity more heavily, preferred the simpler ARMA(0,1) model ($BIC = -1437.29$).

We further assessed promising models — ARMA(0,1), ARMA(1,0), ARMA(1,1), ARMA(2,2), and ARMA(3,3) — based on parameter significance, residual diagnostics using the Ljung-Box test (lag 12), and residual autocorrelation (Box-Pierce up to lag 24).

The ARMA(0,1) model stood out for its simplicity and effectiveness. Its MA(1) coefficient was highly significant. While the Ljung-Box test yielded a p -value of 0.0034, suggesting some residual autocorrelation, the overall fit remained satisfactory. BIC's selection further supports this model.

In contrast, ARMA(1,0) and ARMA(1,1) failed to adequately capture the data's structure. Both showed significant residual autocorrelation ($p < 0.01$), and the AR term in ARMA(1,1) was not significant, undermining their validity.

The ARMA(2,2) model provided a better fit but failed to eliminate autocorrelation. Three of four coefficients were significant, yet the AR(2) term was not. Similarly, ARMA(3,3), though favored by AIC, included a non-significant AR(3) term and residual autocorrelation ($p = 0.0072$), indicating potential overfitting.

We therefore recommend the ARMA(0,1) model. It offers a strong balance between simplicity, interpretability, and performance. The ACF/PACF patterns also support a dominant moving average component with little autoregressive structure. This choice is reinforced by residual diagnostics in Figure 4. The standardized residuals fluctuate randomly around zero, and their ACF shows no significant autocorrelation. The Q-Q plot indicates approximate normality, with minor deviations in the tails, while the histogram confirms a close fit to the normal distribution. Though the Ljung-Box p -values fall below 0.05 for some lags, suggesting mild residual autocorrelation, this does not seriously compromise the model given its parsimony and overall adequacy.

In conclusion, ARMA(0,1) provides the most coherent and statistically justified model, balancing simplicity and fit while aligning with both theoretical expectations and empirical findings. Although the PACF pattern could suggest a strictly positive AR coefficient, the BIC and residual diagnostics consistently indicate that the autoregressive component can be omitted, favoring a purely moving average model with $p = 0$.

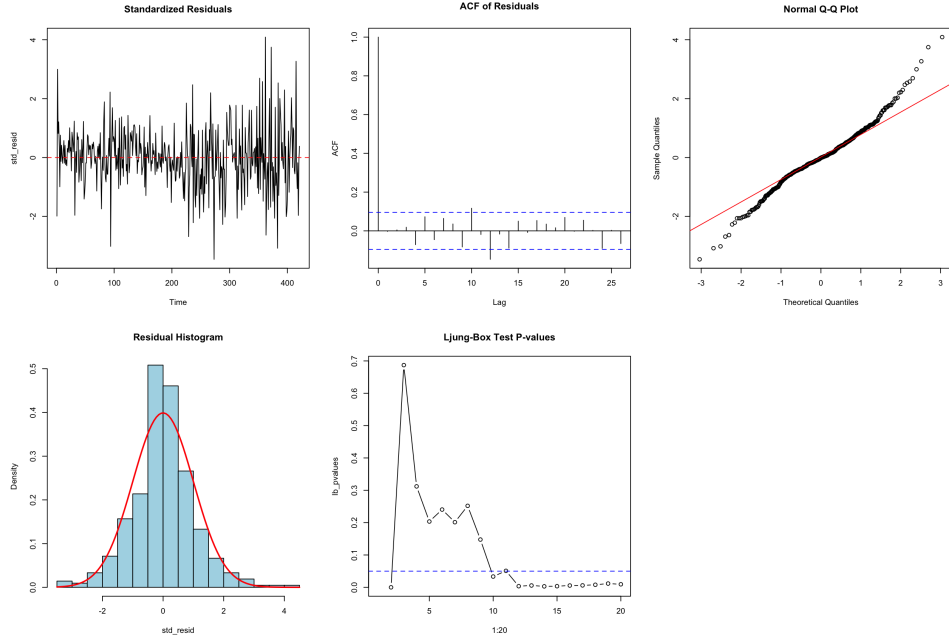


Figure 4: Residual diagnostics

Estimated ARIMA Model

In general, an ARIMA model is specified as:

$$\phi(B)(1 - B)^d \log(X_t) = \theta(B)\varepsilon_t$$

Our analysis led us to identify $d = 1$, $\phi(B) = 1$, and $\theta(B) = 1 - 0.6991B$, corresponding to an **ARIMA(0,1,1)** model applied to the series $\log(X_t)$.

Thus, without accounting for outliers, we obtain the following model:

$$Y_t = \varepsilon_t - 0.6991\varepsilon_{t-1}$$

where $Y_t = (1 - B)^1 \log(X_t) = \log(X_t) - \log(X_{t-1})$.

Part III – Forecasting

Forecast Interval Equation

The $(1 - \alpha)$ confidence intervals for X_{T+1} and X_{T+2} are given by:

$$\begin{cases} IC_{1-\alpha}(X_{T+1}) = \left[\hat{X}_{T+1|T} - z_{1-\alpha/2} \times \hat{\sigma}_\varepsilon; \hat{X}_{T+1|T} + z_{1-\alpha/2} \times \hat{\sigma}_\varepsilon \right] \\ IC_{1-\alpha}(X_{T+2}) = \left[\hat{X}_{T+2|T} - z_{1-\alpha/2} \times \hat{\sigma}_\varepsilon \sqrt{1 + (1 + \hat{\phi} - \hat{\theta})^2}; \hat{X}_{T+2|T} + z_{1-\alpha/2} \times \hat{\sigma}_\varepsilon \sqrt{1 + (1 + \hat{\phi} - \hat{\theta})^2} \right] \end{cases}$$

In this expression, $z_{1-\alpha/2}$ denotes the quantile of order $1 - \alpha/2$ from the standard normal distribution $\mathcal{N}(0, 1)$. The term $\hat{\sigma}_\varepsilon$ corresponds to the empirical standard deviation of the residuals, and is computed as:

$$\hat{\sigma}_\varepsilon = \sqrt{\frac{1}{T} \sum_{t=1}^T \hat{\varepsilon}_t^2}$$

where $\hat{\varepsilon}_t$ denotes the residual at time t , and T is the sample size. And ultimately, $\hat{\phi} = 0$ and $\hat{\theta} = -0.69$.

Assumptions

Assumptions for Constructing the Confidence Region

1. **Convergence of Estimators:** The estimators are consistent, i.e., they converge in probability to the true parameter values as the sample size increases.
2. **Independence and Identical Distribution of Residuals:** The residuals ε_t are i.i.d., ensuring each has the same distribution and is independent of others.
3. **Normality of Residuals:** The residuals follow a normal distribution: $\varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2)$.
4. **Invertibility and Positive Definiteness of the Variance-Covariance Matrix:** The variance-covariance matrix Σ is invertible and positive definite, with $\sigma_\varepsilon^2 > 0$.

Forecast Visualization

The predicted index values for March and April 2025 are 6.253 and 6.257, respectively. The following figure 5 illustrates these individual forecasts along with their corresponding confidence intervals:

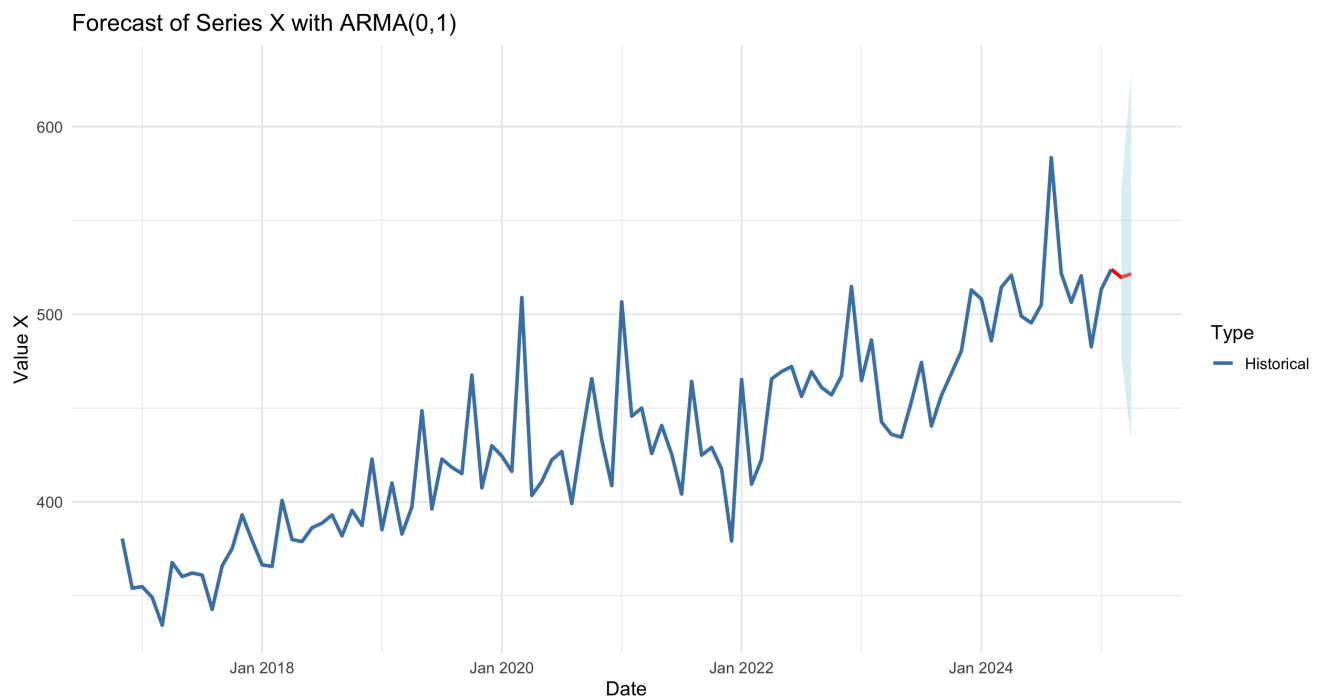


Figure 5: One-step-ahead forecasts and confidence intervals for the pharmaceutical production index in metropolitan France (March and April 2025)

Open Question

Y_{T+1} can improve the prediction of X_{T+1} if Y_t instantaneously causes X_t in the Granger sense. Mathematically, this corresponds to:

$$\hat{X}_{T+1}|\{X_t, Y_t, t \leq T\} \cup \{Y_{T+1}\} \neq \hat{X}_{T+1}|\{X_t, Y_t, t \leq T\}.$$

In other words, Y_{T+1} provides additional information beyond the past data.

To test whether Y_{T+1} improves the prediction of X_{T+1} , we rely on the definition of *instantaneous causality* based on the correlation of innovations.

Let the innovation vector be defined as:

$$\begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix} = \begin{pmatrix} X_t - \hat{X}_t \mid \{X_u, Y_u, u \leq t-1\} \\ Y_t - \hat{Y}_t \mid \{X_u, Y_u, u \leq t-1\} \end{pmatrix}.$$

We then obtain the following equivalence:

$$X_t \text{ does not instantaneously cause } Y_t \iff \varepsilon_{1t} \text{ and } \varepsilon_{2t} \text{ are uncorrelated.}$$

This provides a practical way to test for instantaneous causality.